AN INJECTIVE MODULE WHOSE ASSOCIATED SHEAF ON AN AFFINE SCHEME IS NOT FLASQUE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is 'See above'; the final standard-open restriction failure is the sharper second sentence of the statement, witnessed by U = D(x).

Editorial note. The label lemma-nonflasque names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 3734-3743: X = Spec(A) with A = k[x, y, z_1, z_2, ...]/(x^n z_n), the ring from Properties, Example does-not-work-in-general; I = (x); U = D(x) quasi-compact open; as seen in the referenced Properties example there is a section s of O_X(U) which does not come from an A-module map I^n -> A for any n >= 0
- Context span 3745-3766: alpha: A -> J an embedding into an injective A-module, Q = J/alpha(A) with quotient beta; the claim is that Gamma(X, J~) -> Gamma(U, J~) is not surjective since alpha(s) is not in the image; by contradiction, x in J restricting to alpha(s) over U gives beta(x) in Q restricting to 0 over U, hence I^n beta(x) = 0 for some n (Properties, sections-over-quasi-compact-open-in-affine); this yields phi: I^n -> A, h -> alpha^{-1}(h x), which by the same Properties lemma gives rise to the section s - contradicting the choice of s

Source reference. Original section 01PN: sources/stacks-chow/tags.txt, source lines 2236–2236. Prerequisite pointer invoked at 3784; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01PM: sources/stacks-chow/tags.txt, source lines 2235–2235. Prerequisite pointer invoked at 3805; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Consider the affine scheme $X = \mathrm{Spec}(A)$ where

$$A = k[x, y, z_1, z_2, \ldots]/(x^n z_n)$$

is the ring from Properties, Example 01PN. Set $I = (x) \subset A$. Consider the quasi-compact open $U = D(x)$ of X. We have seen in loc. cit. that there is a section $s \in \mathcal{O}_X(U)$ which does not come from an A-module map $I^n \to A$ for any $n \geq 0$.

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let $\alpha : A \to J$ be the embedding of A into an injective A-module. Let $Q = J/\alpha(A)$ and denote $\beta : J \to Q$ the quotient map. We claim that the map

$$\Gamma(X, \widetilde{J}) \longrightarrow \Gamma(U, \widetilde{J})$$

is not surjective. Namely, we claim that $\alpha(s)$ is not in the image. To see this, we argue by contradiction. So assume that $x \in J$ is an element which restricts to $\alpha(s)$ over U. Then $\beta(x) \in Q$ is an element which restricts to 0 over U. Hence we know that $I^n\beta(x) = 0$ for some n, see Properties, Lemma 01PM. This implies that we get a morphism $\varphi : I^n \to A$, $h \mapsto \alpha^{-1}(hx)$. It is easy to see that this morphism φ gives rise to the section s via the map of Properties, Lemma 01PM which is a contradiction.

Lemma 0.1. *There exists an affine scheme $X = \mathrm{Spec}(A)$ and an injective A-module J such that $\widetilde{J}$ is not a flasque sheaf on X. Even the restriction $\Gamma(X, \widetilde{J}) \to \Gamma(U, \widetilde{J})$ with U a standard open need not be surjective.*

Proof. See above. □